UNIT 0 · THE TOOLBOX · SKILLS CHECK MINI-PROJECT · OPTION 3

The Cold Snap

Computer Science Math 2026–27 · in class Thursday, September 10 · **45 minutes, start to finish**

NAME _____ DATE _____

The job. A weather station logged two weeks of temperatures before it was shut down for a cold snap. The export is a mess, and the station recorded everything in °F — but your report has to be in **°C**. You get the data, you **clean it**, you **convert with one formula**, and you **pick the chart that tells the story of the two weeks.** This is a skills check, not a model: it checks the tools you already have — and nothing you haven't learned yet.

What you receive

ONE CSV · FOURTEEN DAYS

0 The station log

MESSY ON PURPOSE

Three columns: `Date`, `High (°F)`, `Low (°F)`, plus a `Notes` column. Most days are fine. Several are not — that is the point. Expect to find, in no particular order:

- some temperatures typed as text, e.g. `"68F"`
- one **blank day** (the station was down)
- one value of `212` — a data-entry error (nothing here reaches boiling)
- **two readings actually recorded in °C** and mislabelled — the `Notes` column hints at it, but the numbers give it away too
- inconsistent date formats

You do not need VLOOKUP for this. Nothing here is a lookup — that tool belongs to the food tracker next unit. Today is about honest formulas, clean columns, and the right chart.

The build — step by step

BUDGET YOUR 45 MINUTES

STEP	TASK	TIME
1	Set up the sheet and the KDEA header	~5 min
2	Clean the data	~12 min
3	Convert and compute	~13 min
4	Choose and build the chart	~10 min
5	Label check and conclusion	~5 min

1 Set up the sheet

STRUCTURE · KDEA

One tab, top to bottom: a title, a **labelled conversion block**, the data table, a results block, the chart. Every column header carries a **unit** (`°F`, `°C`, `day`). At the top, run the protocol:

K	THE GIVENS: THE LOG, THE CONVERSION RELATIONSHIP
D	THE TWO WEEKS IN °C · THE STORY THE TREND TELLS
E	THE CONVERSION °C = (°F − 32) × 5/9 · AND Δ = HIGH − LOW
A	ONE FORMULA, FILLED DOWN · LABELS ON EVERY COLUMN

Write the conversion as one formula and fill it down. Type the arithmetic by hand even once and the column is no longer a formula.

2 Clean the data

THE PART THAT EARNS THE GRADE

a. **Text to number.** A `"68F"` is text. Strip the letter and convert (`=VALUE( ... )` after trimming the `F`), or retype it. Left-aligned numbers are strings.

b. **Catch the two °C values.** A “high” of `24` in a column of `60`s and `70`s is a sanity-check alarm, not a cold day. Treat them correctly — and say how you knew.

c. **Flag the 212 — do not delete it.** Mark it clearly as an error and keep the evidence. Erasing data isn't cleaning.

d. **Decide about the blank day.** Leave the gap and say so, or mark it “station down.” A blank plotted as `0°C` would lie.

e. **Unify the date format.** Pick one and apply it to every row.

Leave a short **“What I cleaned and why”** note on the sheet. Cleaning is part of the answer.

3 Convert and compute

FORMULAS, NOT NUMBERS

a. **Convert:** `°C = (°F − 32) × 5/9`. Build it once and fill it down. Convert the High and the Low.

b. **Daily swing:** `Δ = High − Low` (in °F, or in °C — one, and label it).

c. Results block: `=MAX` warmest day · `=MIN` coldest day · `=AVERAGE` · `=COUNT` days with valid data.

d. Check yourself: did the average move after you removed the 212 and corrected the two °C rows? If not, one of them is still in the numbers.

4 Choose the chart — and defend it

THE DECISION IS GRADED

The question decides the chart:

- **Line chart** — temperature across the fourteen days. Best when the question is *"how did the trend move over time?"*
- **Column chart** — e.g. the daily swing, Δ, day by day.

Build your pick with a **title, both axis labels, and units on the value axis**, then write one sentence: *why this chart and not the other.* A pie chart of temperatures is not an option — if you were tempted, write down why it fails.

5 Conclusion

ONE OR TWO SENTENCES

Describe the trend, name the coldest and warmest days, and say what would change your reading (the blank day? the flagged 212? the two corrected rows?). A conclusion that only describes the chart is not a conclusion.

Where students trip up

READ THIS BEFORE YOU START

- **Converting text.** `"68F"` won't convert — a text temperature produces `#VALUE!` or a wrong number. Fix the type before the formula.
- **Double conversion.** Running the °F→°C formula on the two rows that were *already* °C turns a fine reading into nonsense. The sanity check is the tool here, not the formula.
- **Deleting the outlier.** Removing the 212 hides a data problem; flagging it proves you found it. Keep the evidence.
- **The blank day plotted as zero.** A missing day that becomes `0°C` invents a cold snap that never happened. Gap it.
- **Line vs. column.** Time-series data wants a line. A pie or a plain column chart of the daily values loses the trend.
- **Averaging High and Low together.** They are two different series. Averaging them into one number erases the story.
- **Unlabelled chart.** No title or no °C on the axis scores below proficient even when the math is perfect.

What you hand in

ONE WORKBOOK

- One tab: title · KDEA header · conversion block · clean data · results block · chart · conclusion.
- A formula in every computed cell — **no typed answers**.
- Every column header carrying its unit; every chart axis labelled.
- Your “What I cleaned and why” note, including the flagged 212.

Scored on the Toolbox rubric: Model Correctness, Formula Honesty, Data Structure, Communication. The tracker — and VLOOKUP — come next.